

Goal

- Examine the use of a variational quantum eigensolver through application to a transverse field Ising model
- Compare the accuracy of a VQE to exact diagonalization solution
- Determine limitations of VQEs

Ising Model

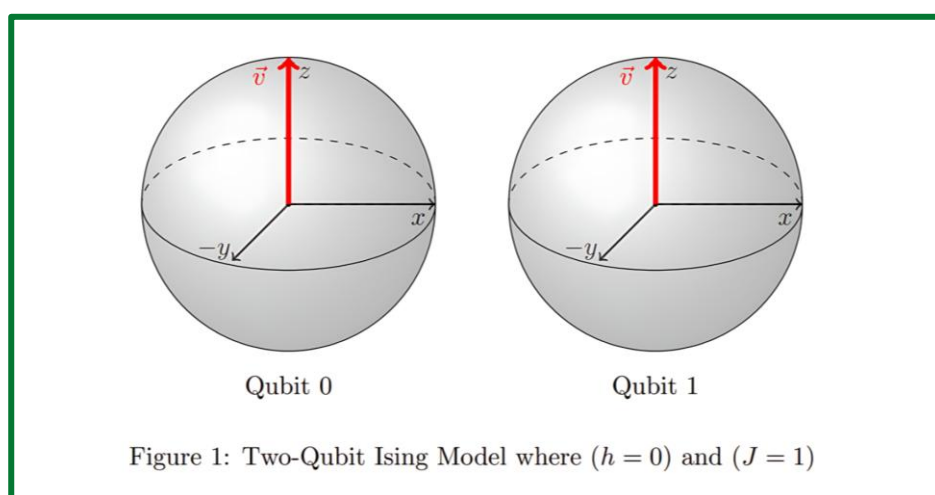


Figure 1: Two-Qubit Ising Model where ($h = 0$) and ($J = 1$)

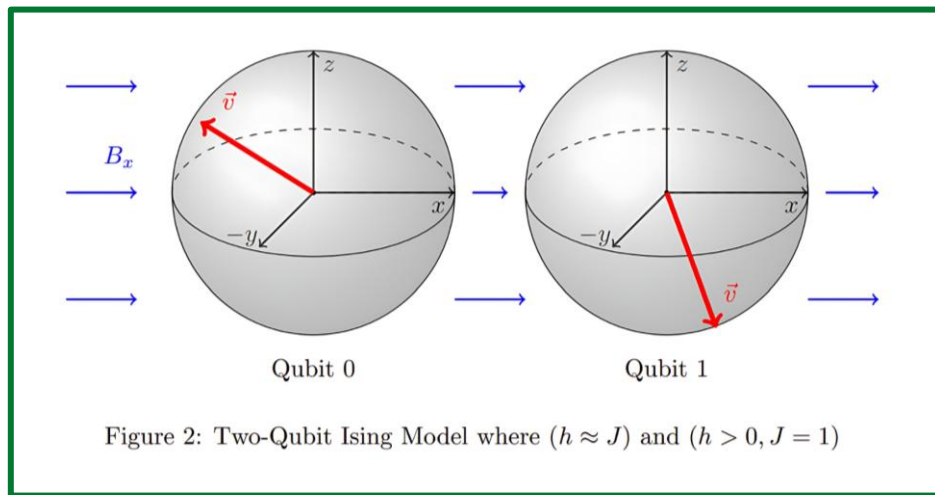


Figure 2: Two-Qubit Ising Model where ($h \approx J$) and ($h > 0, J = 1$)

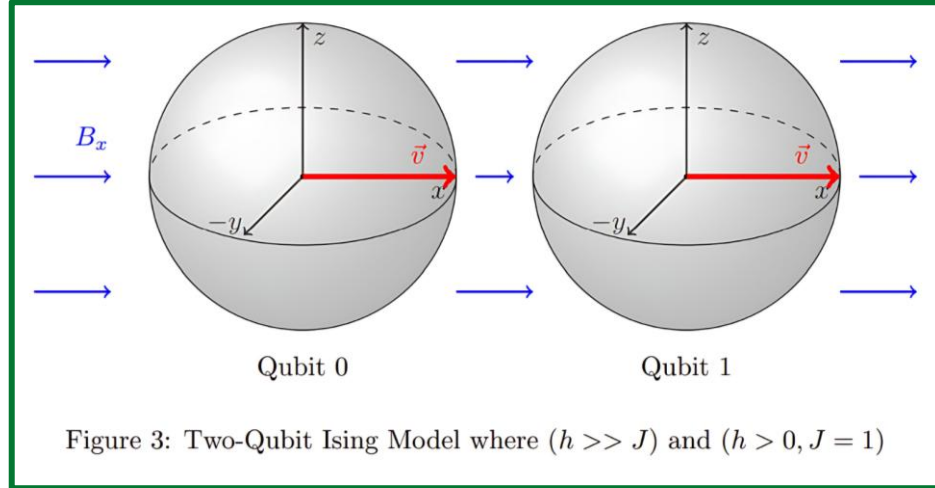


Figure 3: Two-Qubit Ising Model where ($h \gg J$) and ($h > 0, J = 1$)

- Nearest-neighbor interactions
- Isotropic system (equal chance of interaction in any direction)
- Paramagnetic vs. Ferromagnetic regions
- Phase transition between regions

Classical vs. Quantum

$$T_{\text{classical}}(N) = O(N^3)$$

$$T_{\text{quantum}}(N) = O(\text{poly}(N))$$

Approximate runtime for classical and VQE methods

4 Qubit Ising Model

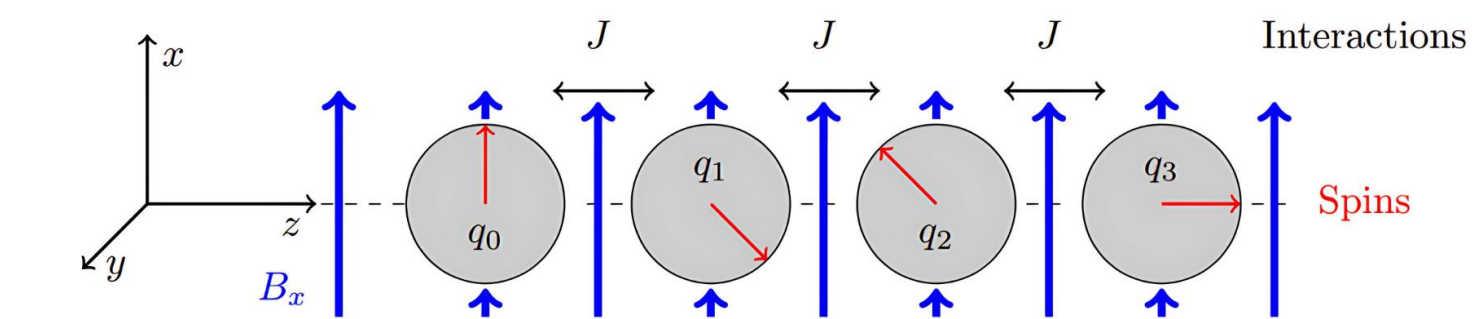
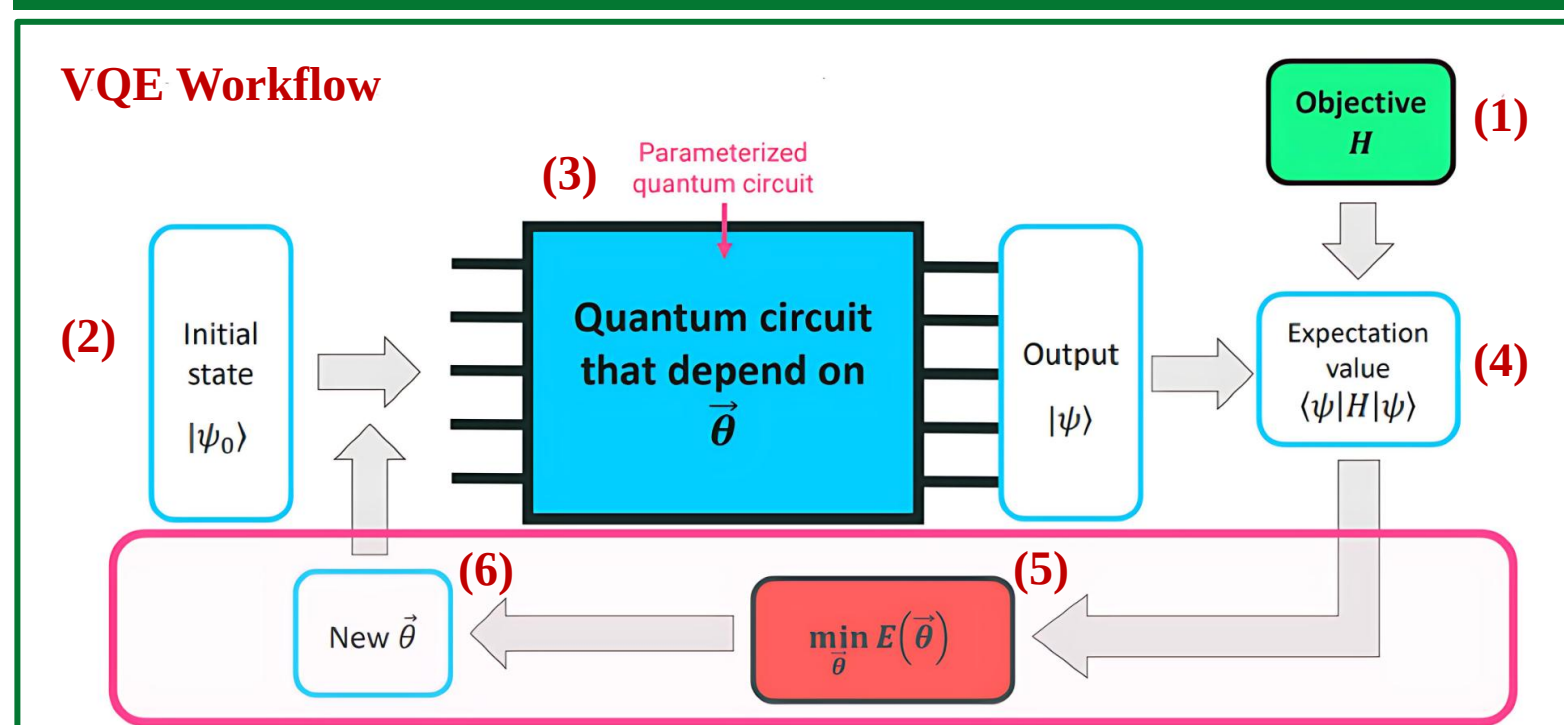
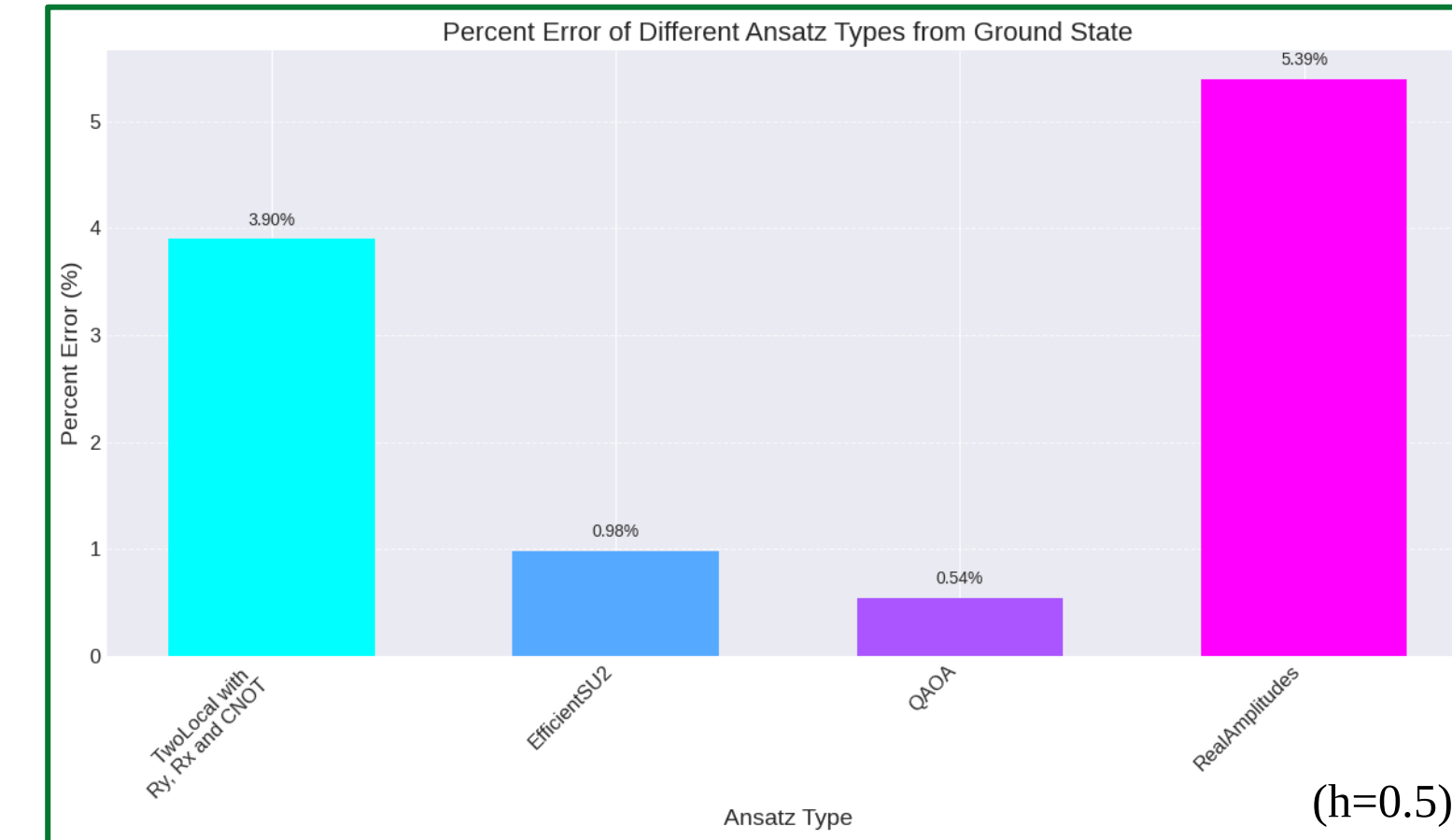
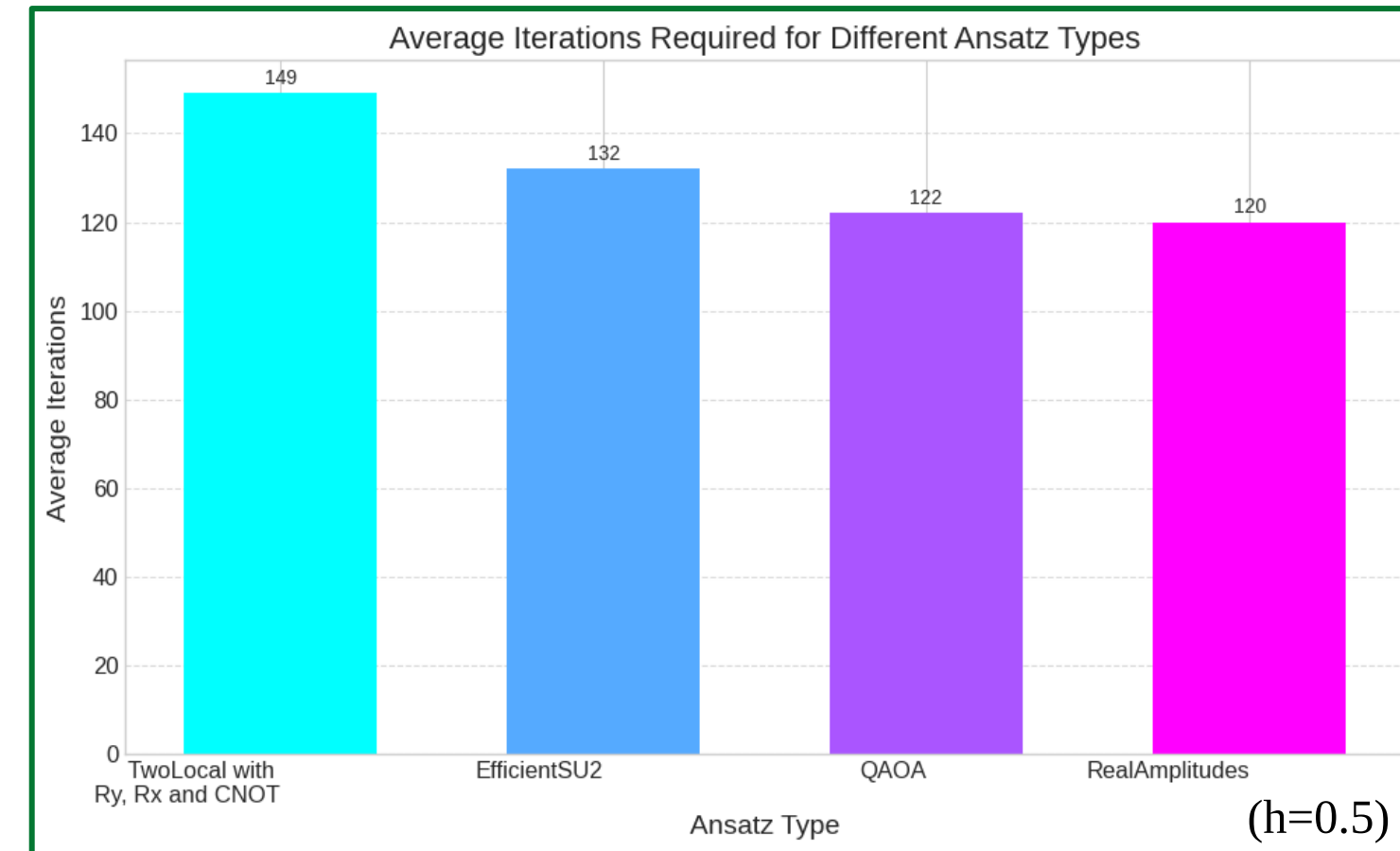


Figure 4: Four Qubit Ising Model

Variational Quantum Eigensolver (VQE)



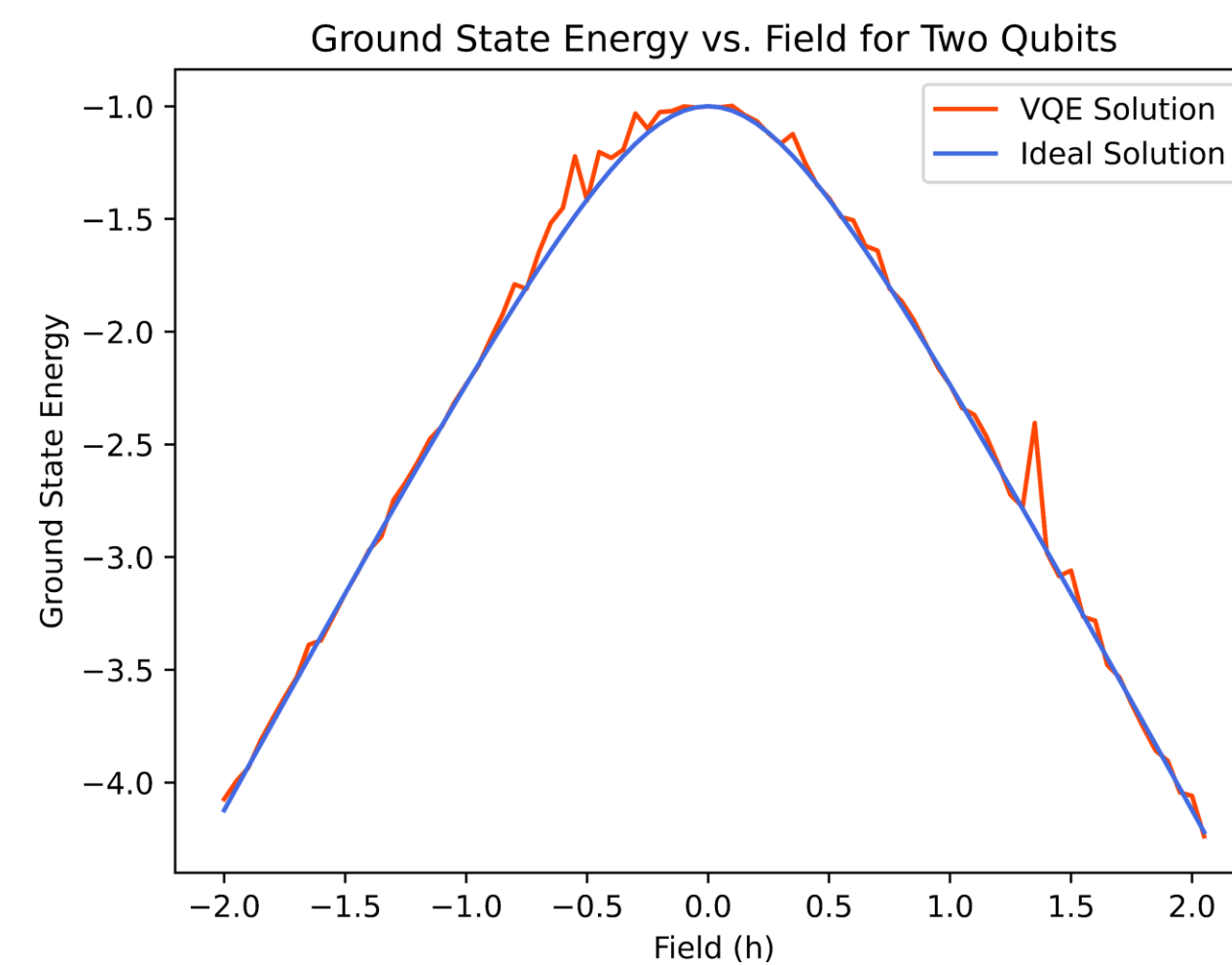
Ansatz Comparison (16 Parameters)



2 Qubit Model

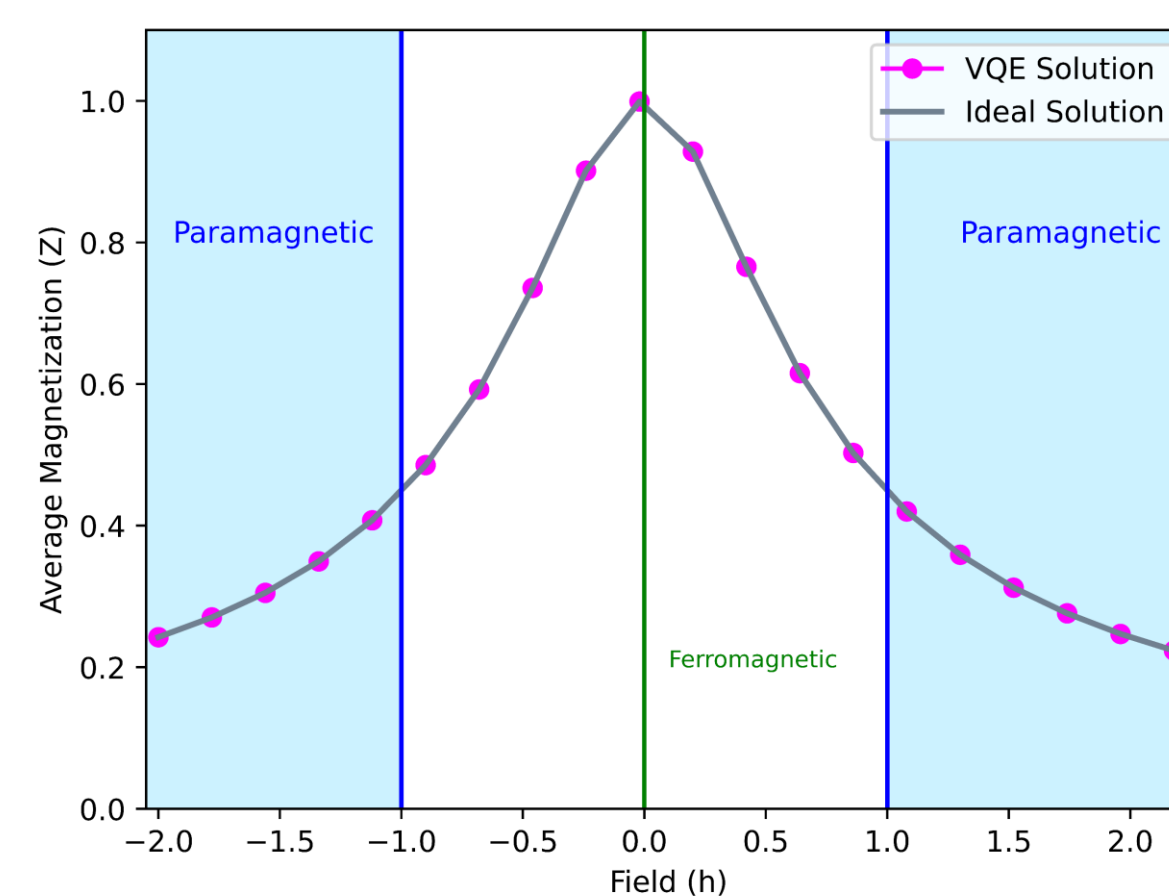
- The Hamiltonian
- Comparison of VQE and Ideal Solution
- Case 1: Ground State Energy

$$\bar{H} = -J(Z_1 Z_2) + h(X_1 + X_2)$$



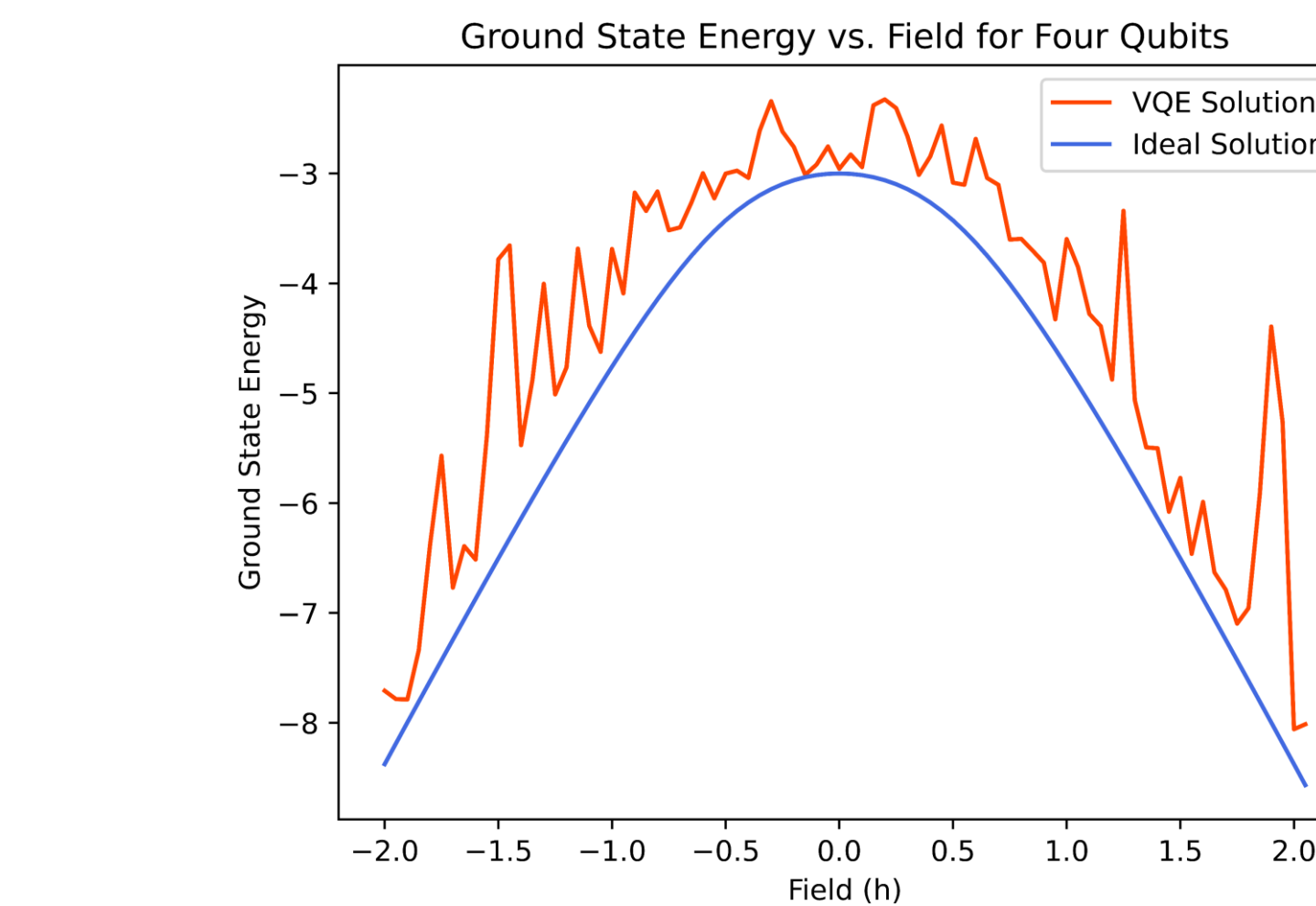
- Case 2: Magnetization

Average Magnetization (Z) vs. Field (h) for Two Qubit System



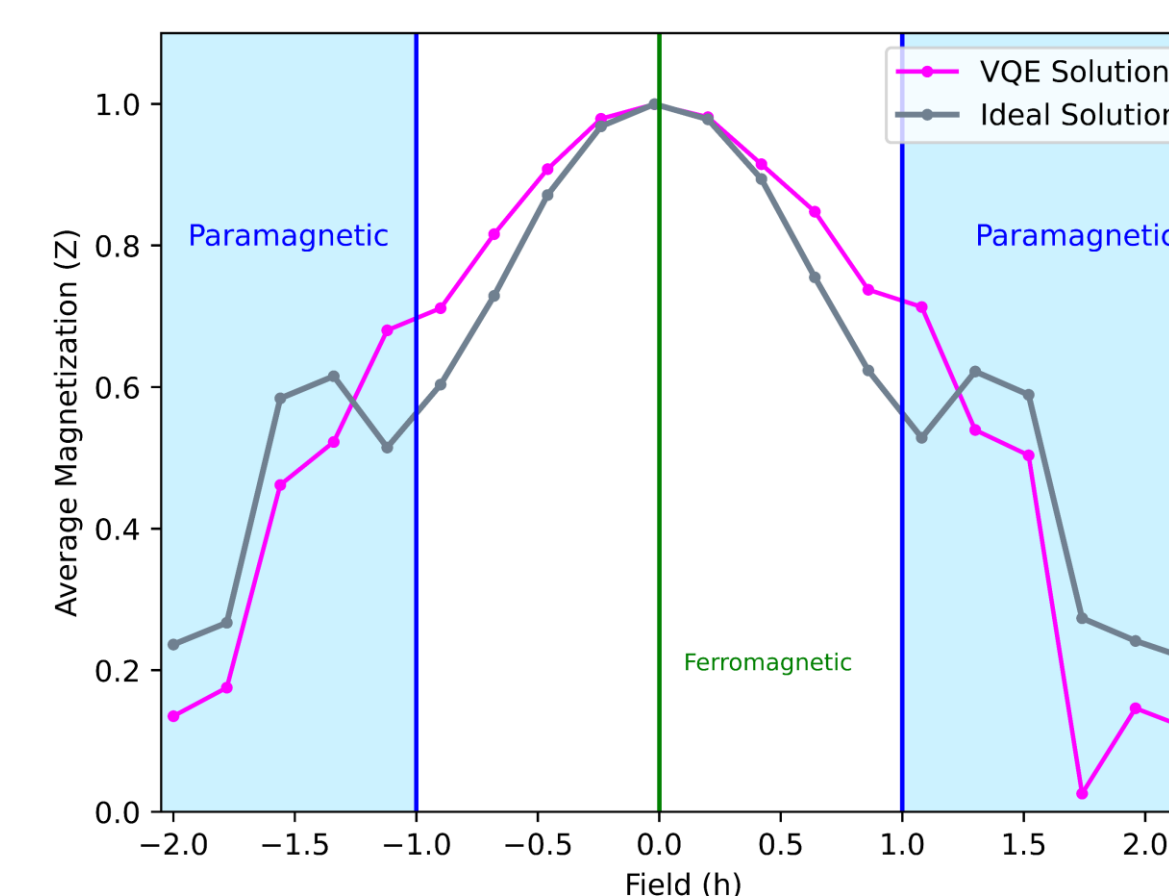
4 Qubit Model

- The Hamiltonian
- Comparison of VQE and Ideal Solution
- Case 1: Ground State Energy

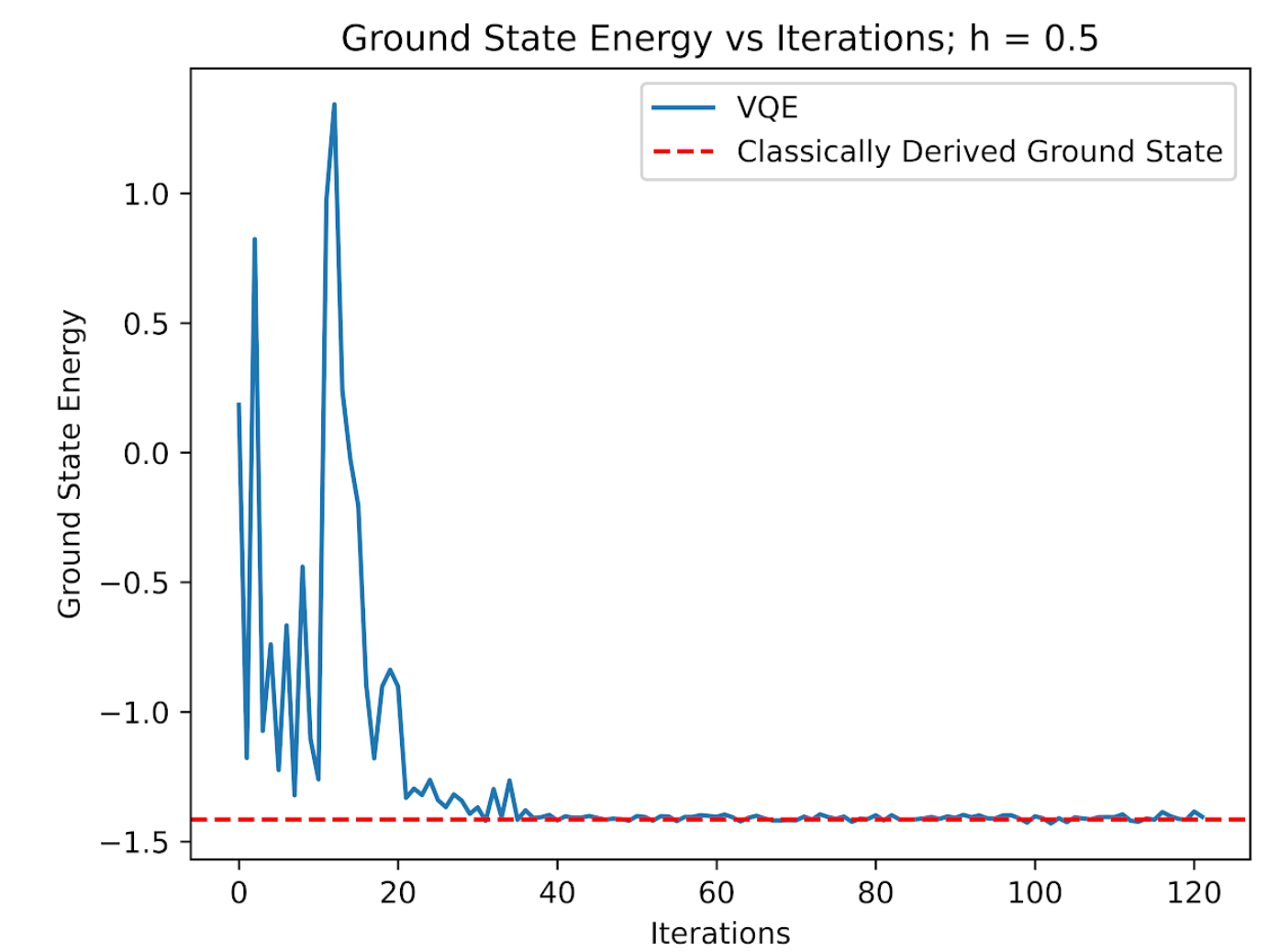


- Case 2: Magnetization

Average Magnetization (Z) vs. Field (h) for Four Qubit System



Ground State Convergence



Analysis

- More complex system requires more computational power
- Symmetric graphs about $h=0$. Error present varies.
- Difficulty converging around phase transition state
- VQE limitations: barren plateaus
- Parameter optimization as circuit depth increases

Future Research

- Include simulation of noise to determine potential impact of using a VQE on a quantum computer
- Simulate different spin models with other interactions

References

- IBM (2025) *Quantum approximate optimization algorithm*, IBM Quantum Documentation. Available at: <https://quantum.cloud.ibm.com/docs/en/tutorials/quantum-approximate-optimization-algorithm> (Accessed: 23 June 2025).
- Watrous, J. (2025a) *Basics of quantum information*, IBM Quantum Learning. Available at: <https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information> (Accessed: 23 June 2025).
- Watrous, J. (2025b) *Fundamentals of quantum algorithms*, IBM Quantum Learning. Available at: <https://quantum.cloud.ibm.com/learning/en/courses/fundamentals-of-quantum-algorithms> (Accessed: 23 June 2025).
- Watrous, J. (2025c) *Variational algorithms*, IBM Quantum Learning. Available at: <https://quantum.cloud.ibm.com/learning/en/courses/variational-algorithm-design/variational-algorithms> (Accessed: 23 June 2025).
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